Discussion Notes

IYPT 2026 — Problem #1: Ring Fountain

POINT #1 — Cavity Mechanics Absent from the Theoretical Model

Question: On slide 6 you use momentum conservation to find the velocity of the ring just after impact; on slide 7 you use continuity to find the instantaneous jet velocity; and on slide 8 you use energy conservation to find the maximum height, assuming the jet starts at that velocity. You state throughout the presentation that the cavity collapse is the physical mechanism driving the jet. Yet nowhere in this energy budget is there any term accounting for the energy added to the jet by the cavity collapse. Is that correct?

Personal opinion: This is a meaningful gap. The cavity collapse is presented as the driver of the jet, yet it contributes nothing to the energy balance in the model. A jet speed estimated purely from the ring's initial momentum will therefore systematically underpredict the true jet height, and the discrepancy will grow with ring momentum. The energy input from the collapsing cavity should either be modeled explicitly or at least acknowledged as a source of systematic error.

POINT #2 — Incorrect Added Mass Treatment

Question: On slide 6 you invoke Archimedes' principle to justify that the volume of water carried along with the ring as added mass equals the volume of the ring itself. Archimedes' principle concerns buoyant forces on objects *at rest* due to static pressure differences — it does not apply to a moving object undergoing an impulsive collision with a free surface. How is it relevant here?

Question: More broadly, the added mass of a body moving through a fluid depends on the geometry of the object and is generally not equal to its displaced volume. For a sphere, for instance, the added mass corresponds to half its volume; values differ further for other shapes. Did you attempt to find a more precise estimate for your specific ring geometry?

Personal opinion: This affects the model at a foundational step. Using an incorrect added mass coefficient corrupts the momentum conservation calculation, propagating the error into every subsequent quantity. A more precise estimate — even a literature value for a toroidal or annular geometry — would likely improve the accuracy of the predicted jet velocities and heights by a non-trivial factor.

POINT #3 — Efficiency Factor *k* — Fitted or Independent?

Question: On slide 8 you introduce an efficiency factor k, described as "typically smaller than 1", to account for energy losses to waves, turbulence, splash, and similar effects — none of which are explicitly modeled. With what value of k were the results in the tables obtained? Was k determined by fitting to the experimental data, or estimated independently before comparison?

Personal opinion: If k is fitted to the experimental data, the model loses all predictive power and the comparison between theory and experiment becomes circular. The value

of k and the method by which it was determined should be stated clearly. If it was fitted, it should be presented as an empirical correction rather than a model prediction.

POINT #4 — Height Measurement Definition

Question: How did you experimentally define the maximum height of the ring fountain — was it the highest point reached by an individual droplet, or the highest point the unified jet reached before collapsing? Individual droplets clearly travel higher than the main jet, so which did you choose to measure?

Personal opinion: The correct experimental observable should be the maximum height of the unified jet. By momentum conservation, droplets that travel above the main jet carry momentum away from it, reducing the jet height slightly below what the initial jet speed alone would predict. Since the theoretical model on slide 8 is built around jet speed, the jet height is the appropriate quantity to compare against. Using droplet height instead would introduce a systematic overestimate.

POINT #5 — Growing Deviation — Unmodeled Dissipation

Question: For the increasing discrepancy between experimental and theoretical values at larger jump heights on slide 15, you attribute the cause to "errors during the falling of the ring." What other physical effects do you think might be contributing?

Personal opinion: Aerodynamic drag during the fall and hydrodynamic drag during water entry both scale with velocity, so they would naturally produce a deviation that grows with height. Additionally, the acoustic and surface-wave energy radiated at impact increases at higher impact speeds, representing further losses not captured in the model. If these dissipation mechanisms were modeled — even approximately — the trend could be explained quantitatively rather than attributed to measurement error, and the need for a catch-all efficiency factor would be reduced.

POINT #6 — No Error Analysis

Question: There are no error bars on any of the diagrams, and no discussion of measurement uncertainty anywhere in the presentation. Were all heights measured using Tracker? If so, what do you estimate the measurement error to be, and why is it not reported anywhere?

Personal opinion: The absence of uncertainty quantification undermines the credibility of all experimental results. Without error bars it is impossible to judge whether the agreement or disagreement between theory and experiment is statistically meaningful. The uncertainty inherent in Tracker measurements — arising from pixel resolution, frame rate, and manual tracking — should be estimated and reported.

POINT #7 — Unused Dimensionless Parameters

Question: You define several dimensionless parameters during the presentation but do not appear to use them anywhere in the analysis. What role do they play?

Personal opinion: If these parameters are not used in the theoretical model, not plotted against experimental data, and not used to characterise the flow regime, their inclusion creates unnecessary clutter without contributing to the scientific argument. Either they should be used — for example, to non-dimensionalise the height measurements or to define the regime in which the experiment operates — or they should be removed.

POINT #8 — Conclusions Already Implicit in the Model

Your first two conclusions on slide 20 follow directly from the theoretical model and do not require experimental confirmation to state. It may be worth reflecting on whether they add anything beyond what the model already predicts, or whether they would be better framed as validations of the model's assumptions rather than independent findings.